The Adaptation–Embodied Carbon Paradox: A Lifetime-Aware Theory of Continual Learning on Wear-Limited Edge Hardware

Extended Version

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Abstract

Green-AI work usually minimizes *operational* edge-inference energy, yet tiny AIoT devices often incur most life-cycle emissions as *embodied* carbon amortized over physical lifetime. Continual Learning (CL) is needed under concept drift, but every on-device update spends finite non-volatile-memory (NVM) endurance. We identify the *Adaptation–Embodied Carbon Paradox*: adaptation preserves accuracy while accelerating wear-out, replacement, and embodied-carbon emissions. We therefore treat learning algorithms as physical agents constrained by carbon and endurance, not merely software objectives, and recast stability–plasticity as a *stability–plasticity–endurance trilemma*. We prove an embodied-carbon amortization lemma, a no-free-lifetime bound, a Lyapunov near-optimality theorem for *Stability-Controlled Lifetime-Aware Continual Learning* (SC-LACL), a deterministic (non-asymptotic) greedy wear-leveling peak bound, and a lifetime-and-carbon bound. We also propose the Total Carbon-Lifetime-Delay Product (tCLDP). Results are grounded simulations using counted per-cell P/E wear and datasheet endurance/energy parameters, not yet hardware lifetime tests. Across three drift streams and seven baselines, wear-agnostic CL exhausts memory on two streams and is throttled by harvested energy on the third, whereas SC-LACL nearly matches accuracy with 13–30× fewer NVM writes, survives on one device, and attains the best tCLDP; a 19× sparsity sweep suggests most plasticity is redundant wear, and a static shadow-price baseline matches SC-LACL until endurance turns severely scarce, where adaptivity cuts device count 4.6×.

CCS Concepts

• **Computing methodologies** → **Lifelong machine learning**; • **Hardware** → *Non-volatile memory*; • **Social and professional topics** → *Sustainability*; • **Theory of computation** → Stochastic control and optimization.

Keywords

sustainable computing, embodied carbon, continual learning, NVM endurance, Lyapunov optimization, energy harvesting, edge AI

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1 Introduction

The Information and Communication Technology sector emits an estimated 2.1–3.9% of global greenhouse gases [14]. In consumer electronics, IoT sensors, and edge-AI devices, embodied carbon from manufacturing, packaging, and transport often dominates life-cycle emissions at 70–90% [10, 17]. Since this fixed cost is amortized over lifetime, device longevity directly lowers annualized emissions; reported cradle-to-gate footprints range from ≈0.52 kg CO_2e (Raspberry Pi Pico) to ≈2.59 kg CO_2e (Coral Dev Board Micro), while IoT deployments already exceed 16.6 billion devices [8, 10].

Edge data is non-stationary, so static models degrade under concept drift [15]; CL restores accuracy by updating weights in place [21, 24, 25, 32]. But microcontrollers have only 128–512 KB SRAM, so training state spills to Flash or emerging NVM, where finite P/E endurance (10^4–10^5 for commodity Flash; $\sim 10^6$ for flash-replacement embedded STT-MRAM) and write energy (e.g., 16.82 pJ/bit for PCM) matter [7, 19, 23]. This exposes the **Adaptation–Embodied Carbon Paradox** (Fig. 1): writes sustain accuracy but shorten physical life, and replacement embodied carbon can outweigh operational savings. We use *paradox* in the self-defeating-optimization sense: the writes that sustain accuracy are exactly what can raise *net* life-cycle carbon above a static baseline (Lemma 1, Table 1).

Contributions. (1) We formalize the paradox and make device lifetime endogenous to the learning algorithm, adding a physical endurance axis to stability–plasticity (Sec. 3). (2) We prove carbon amortization, no-free-lifetime, Lyapunov near-optimality, a deterministic greedy wear-leveling peak bound, and lifetime/carbon bounds (Secs. 3–4). (3) We propose tCLDP and validate against seven baselines on three drift streams, including PRICE, a new static shadow-price baseline that isolates when adaptive control matters (Secs. 5–6).

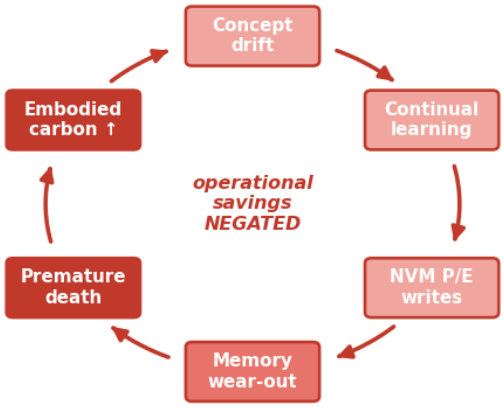


Figure 1: The Adaptation–Embodied Carbon Paradox: plasticity drives NVM wear, premature replacement, and embodied carbon that can offset operational savings.

2 Related Work

Carbon-aware computing. ACT, CORDOBA, CarbonClarity, and spatio-temporal accounting show that embodied carbon dominates and motivate CDP/tCDP metrics [11–13, 16, 38], but treat lifetime as a fixed *input*; we make it an output of learning. FlexiFlow [28] and COFFEE [35] are concurrent and complementary: they fix hardware/lifetime at *design time*, whereas SC-LACL fixes hardware and controls learning *at runtime*. **Continual learning.** EWC, LwF, Experience Replay, and ADWIN-like drift detectors optimize accuracy or forgetting, not write endurance or embodied carbon [3, 15, 21, 24, 32]. Edge and fog learning frameworks [4, 22, 29, 33] likewise target accuracy, latency, energy, or privacy rather than device wear. **Energy harvesting and intermittent computing.** Checkpointing and Lyapunov-stabilized harvesting CL manage *joules*, not write cycles [27, 31]. **Endurance-aware hardware/training.** NVM training, bit-flip-robust methods, approximate flash, memory-light on-device training, and HDC reduce update cost [1, 5, 7, 20, 25, 36, 37]; SC-LACL adds near-optimality/lifetime bounds.

3 The Paradox and the Trilemma

A renewable-powered node's operational carbon $\to$ 0, so total carbon is dominated by embodied carbon C_{emb} amortized over its mean time to failure. For a deployment horizon T, define

$$N_T(M) = \left\lceil \frac{T}{M} \right\rceil, \qquad C_{\text{total}}(M) \approx C_{\text{emb}} N_T(M), \\ M = \text{MTTF}(\rho) = \frac{E_{\text{cyc}}}{\beta\rho}. \tag{1}$$

Here M is mean time to failure, N_T is the number of required devices, E_{cyc} is per-cell endurance, ρ is the plasticity (P/E) rate, and β is write amplification. Thus MTTF is inversely proportional to plasticity: the learner controls its own lifetime.

Lemma 1 (Embodied-carbon amortization). *Let*

$$g(M) = \frac{C_{emb}}{T} \left\lceil \frac{T}{M} \right\rceil \tag{2}$$

be annualized embodied carbon for a device with MTTF = M. Then: (i) $g(M)$ is non-increasing in M; (ii) $g(M) = C_{emb}/T$ for every $M \geq T$; and (iii) among one-device, carbon-optimal policies, the most plastic sustainable target is $M = T$, equivalently $\bar{w} = E_{cyc}/T$.

Proof. The device count $\lceil T/M \rceil$ is a non-increasing step function. It equals 1 when $M \geq T$ and is at least 2 when $M < T$, proving (i) and (ii). Since $M = E_{\text{cyc}}/(\beta\rho)$ decreases as the write rate ρ increases, the largest admissible plasticity rate that still uses one device is obtained at the boundary $M = T$, giving the wear budget $\bar{w} = E_{\text{cyc}}/T$. □

The trilemma. Classical CL trades *stability* (retaining knowledge) against *plasticity* (acquiring it). On wear-limited NVM a third, physical axis is unavoidable: every act of plasticity spends finite endurance. We pose adaptation as constrained control—maximize time-average accuracy subject to (i) an energy buffer that never browns out and (ii) a wear rate $\leq \bar{w}$.

Proposition 1 (No-free-lifetime bound). *Suppose any learner that tracks drift rate δ at time-average accuracy $\geq \alpha$ must issue writes at rate at least $\rho_{\min} = \rho_{\min}(\delta, \alpha)$. Then*

$$MTTF \leq \frac{E_{cyc}}{\beta\rho_{\min}}. \tag{3}$$

A one-device target horizon T is feasible only if

$$\beta\rho_{\min} \leq \frac{E_{cyc}}{T}, \tag{4}$$

and the corresponding update energy is covered by harvested power.

Proof. Equation (1) shows that MTTF = $E_{\text{cyc}}/(\beta\rho)$ is decreasing in ρ. Substituting the necessary rate $\rho \geq \rho_{\min}$ gives the upper bound. By Lemma 1, one-device operation over horizon T requires MTTF $\geq T$; rearranging gives the feasibility condition. □

Proposition 1 is a conditional limit: the paradox cannot be escaped by cleverness alone, only by reducing the writes required per unit of retained accuracy. This motivates SC-LACL.

4 SC-LACL: A Controller for the Trilemma

We build on Lyapunov-stabilized energy-harvesting CL [27], which throttles training from the residual energy state but accounts only for joules. We introduce the missing axis via an **Endurance-Aware Carbon Virtual Queue**

$$Q_Z(t{+}1) = \max\left\{ Q_Z(t) + \frac{w(t)}{\bar{w}} - 1, 0 \right\}, \tag{5}$$

which tracks normalized wear above the ideal linear trajectory of Lemma 1, alongside data (Q_D) and energy-deficit (Q_E) queues. With $L(t) = \frac{1}{2}\sum_j Q_j(t)^2$, SC-LACL minimizes, each slot, the drift-plus-penalty bound

$$V\, p(t) + \sum_j Q_j(t)\Delta_j(t), \tag{6}$$

where $p(t) = \text{loss}(t) + \text{operational carbon}(t)$ and $\Delta_j(t)$ is the increment of queue j. The parameter V trades accuracy against queue stability (Fig. 2).

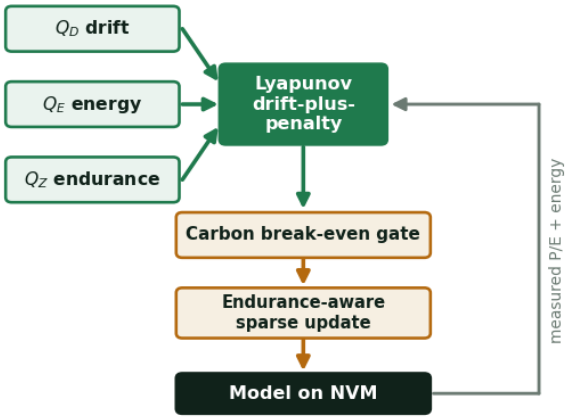


Figure 2: SC-LACL: three virtual queues feed a Lyapunov controller that gates updates (carbon break-even) and applies endurance-aware, wear-leveled sparse updates, closing the loop with measured P/E and energy.

Theorem 1 (Near-optimality; adapts [26]). *Assume i.i.d. random events, a penalty depending only on the current event and action, every queue increment bounded by σ, and data, energy, and wear constraints Slater-feasible with slack $\epsilon > 0$. Let $p^\star$ be the optimal time-average penalty, $p_{\max}$ an upper bound on $p(t) \geq 0$, and $B = \frac{3}{2}\sigma^2$. For every horizon T,*

$$\frac{1}{T}\sum_{t<T}\mathbb{E}[p(t)] \leq p^\star + \frac{B}{V}, \tag{7}$$

$$\frac{1}{T}\sum_{t<T}\sum_{j}\mathbb{E}[Q_j(t)] \leq \frac{B + V p_{\max}}{\epsilon}. \tag{8}$$

Thus the optimality gap is $O(1/V)$ and the average backlog is $O(V)$.

Proof. The queue update and $\max[x,0]^2 \leq x^2$ give the one-step drift bound

$$\Delta(t) \leq B + \sum_j Q_j(t)\mathbb{E}[a_j(t) - b_j(t) \mid \mathbf{Q}(t)]. \tag{9}$$

SC-LACL minimizes the bound on $\Delta(t) + V\mathbb{E}[p(t) \mid \mathbf{Q}(t)]$ over feasible actions, so this quantity is no larger than under any stationary randomized policy. The optimal one (penalty $p^\star$, no expected queue growth) gives $\Delta(t) + V\mathbb{E}[p(t) \mid \mathbf{Q}(t)] \leq B + Vp^\star$; an ϵ-slack one gives

$$\Delta(t) + V\mathbb{E}[p(t) \mid \mathbf{Q}(t)] \leq B + Vp_{\max} - \epsilon\sum_j Q_j(t). \tag{10}$$

Taking expectations, summing over $t < T$, and telescoping $L(t)$ with $L(0) = 0$ and $L(T) \geq 0$, the first bound divided by VT yields (7); the second, with $p(t) \geq 0$, divided by ϵT yields (8). □

Proposition 2 (Greedy wear-leveling bound). *Consider a pool of n physically interchangeable cells with equal initial wear. If every write, of size at most $\delta_{\max}$, is routed to a cell currently attaining $\min_i w_i(t)$ (ties broken arbitrarily), then for all t,*

$$\max_i w_i(t) \;\leq\; \frac{1}{n}\sum_i w_i(t) \;+\; \delta_{\max}. \tag{11}$$

Proof. By induction on write count. At t=0 all w_i=0, so $\max_i w_i - \min_i w_i = 0 \leq \delta_{\max}$. Suppose this spread bound holds before some write. The write lands on a cell at the current minimum m, raising it to at most $m + \delta_{\max}$; every other cell is already $\geq m$ and unchanged, so the new minimum is still $\geq m$ and the new maximum is at most $m + \delta_{\max}$, so the spread remains $\leq \delta_{\max}$. Since $\min_i w_i(t) \leq \mathrm{mean}_i w_i(t)$ always, $\max_i w_i(t) \leq \min_i w_i(t) + \delta_{\max} \leq \mathrm{mean}_i w_i(t) + \delta_{\max}$. □

Theorem 2 (Lifetime and carbon bound). *Under Theorem 1, the wear queue Q_Z is mean-rate stable, and*

$$\overline{w}_T = \frac{1}{T}\sum_{t<T}\mathbb{E}[w(t)] \leq \bar{w}\Big(1 + \frac{\mathbb{E}[Q_Z(T)]}{T}\Big). \tag{12}$$

Under greedy wear-leveling (Prop. 2), peak per-cell wear exceeds the pool mean by at most $\delta_{\max}$ (a single write's worth), so the expected peak cell wear at T is at most E_{cyc} whenever $E_{cyc} \geq \overline{w}_T \cdot T + \delta_{\max}$. If $\max_i w_i(T) \leq E_{cyc}$ almost surely, its embodied carbon is the one-device optimum of Lemma 1.

Proof. Summing the queue update over $t < T$ gives the displayed bound, whose excess term $\mathbb{E}[Q_Z(T)]/T$ vanishes by mean-rate stability. With zero initial wear, $\mathbb{E}[\mathrm{mean}_i w_i(T)] = \overline{w}_T T \leq E_{\mathrm{cyc}} - \delta_{\max}$ by hypothesis. By Prop. 2, the peak cell satisfies $\mathbb{E}[\max_i w_i(T)] \leq E_{\mathrm{cyc}}$, so the hottest cell remains within the endurance limit in expectation. The carbon claim follows from Lemma 1. □

Remark. Prop. 2 assumes physically interchangeable cells and greedy routing within a wear-leveling pool; SC-LACL's allocation only biases writes toward less-worn cells, so the bound describes an idealized router rather than our simulation, and it does not hold across arbitrary hardware layouts (e.g., cells split across independent dies or planes are not interchangeable in this sense, and a physical flash-translation layer may not honor the greedy assignment exactly). Within scope, the $\delta_{\max}$ margin is a fixed constant, not a vanishing $o(1)$ term: it is negligible relative to the slack $E_{\mathrm{cyc}} - \beta\rho_{\min}T$ in the slack regime ($E_{\mathrm{cyc}}/T \gg \beta\rho_{\min}$), where leveling is empirically inactive (Sec. 6), and matters increasingly as $E_{\mathrm{cyc}}/T \to \beta\rho_{\min}$.

Proposition 3 (Carbon break-even rule). *A candidate update that reduces loss by $\Delta\ell$, consumes energy Δe, and spends normalized wear $\Delta w/\bar{w}$ is accepted, ignoring Q_D and operational carbon, exactly when*

$$V\Delta\ell \geq Q_E\Delta e + Q_Z\frac{\Delta w}{\bar{w}}. \tag{13}$$

Proof. Relative to skipping the update, adapting changes the per-slot bound by

$$-V\Delta\ell + Q_E\Delta e + Q_Z\frac{\Delta w}{\bar{w}}. \tag{14}$$

The minimizer accepts the update iff this change is non-positive, which is the displayed inequality. □

As wear runs ahead of budget ($Q_Z \uparrow$), the accuracy bar to justify a write rises, and learning becomes carbon-budgeted decision-making.

Endurance-aware gradient allocation. Each update applies only the top-k highest-magnitude gradients, biased toward the least-worn cells via per-cell counters: algorithmic wear-leveling independent of the flash-translation layer.

5 The tCLDP Metric

Energy-Delay Product and carbon-delay variants (CDP, tCDP) amortize embodied carbon over a *fixed* lifetime [12, 13]. We propose the **Total Carbon-Lifetime-Delay Product (tCLDP)**, making MTTF = $f(\rho)$ from (1) explicit:

$$\text{tCLDP} = \frac{g(M)}{\bar{\alpha}} = \frac{N_T(M)\, C_{\text{emb}}/T}{\bar{\alpha}}, \tag{15}$$

in kg CO_2e·yr^{-1} per unit accuracy, with $g(M)$ from Lemma 1 and $\bar{\alpha}$ the time-average accuracy (inference delay is identical across policies here and omitted); dividing by $\bar{\alpha}$ prevents winning by writes alone (Static: 1.53 vs. SC-LACL's 1.26, Table 1). Validating tCLDP itself against ground truth — correlating it with measured MTTF and audited cradle-to-gate carbon rather than the datasheet/simulation inputs used here — is open; CarbonClarity's [11] and COFFEE's [35] uncertainty-aware carbon models are the natural first step.

6 Experiments

Setup and scope. A small MLP learns online on ELEC2 [18], a 40k Covertype block [6], and SEA [34]. Accuracy is prequential; NVM wear is counted from actual per-cell P/E operations (E_{cyc}=10^4, seasonal solar trace, supercapacitor). Endurance exhaustion triggers replacement and one 2.59 kg embodied-carbon unit; the horizon T is each stream's length at one sample per 30 min (2.6 yr ELEC2, 2.3 yr Covertype and SEA). We compare Static, Naive, Periodic, EWC [21], Experience Replay [32], LwF [24], and PRICE (a static shadow-price baseline, detailed below) with SC-LACL (V=12, top-k=0.18, $\bar{w}$=$0.85E_{\text{cyc}}/T$), over 5 seeds; baselines train 6 epochs/day, SC-LACL 1–4 scaled by drift. The simulated controller applies the break-even test of Prop. 3 with Q_E and Q_Z, estimating $\Delta\ell$ from recent accuracy in place of Q_D; Theorems 1–2 describe the idealized controller. Every policy trains only as many epochs as its energy buffer and harvest can pay for. These are grounded simulations using datasheet endurance/energy constants, not hardware lifetime tests: real P/E endurance is temperature- and pattern-sensitive, write amplification varies with the flash-translation layer, and wear-leveling here is algorithmic (Sec. 4), not validated against a physical FTL.

Main results. Accuracy-oriented CL is wear-agnostic: on ELEC2 and SEA, dense writes exhaust memory at 1.6 yr, require two devices, and give the worst tCLDP (≈2.5–2.6; Table 1); on Covertype, the larger model's dense updates are instead throttled by harvested energy (one device, 30M writes). SC-LACL stays within 1.5 accuracy points of the best baseline while issuing 13–30× fewer writes (Fig. 3; SEA: 5.65 vs. 0.19M), surviving on one device and attaining the best tCLDP on all three streams, narrowly on Covertype (1.310±0.004 vs. 1.319±0.005; Fig. 4).

Why it works. The sparsity sweep (Fig. 5) shows accuracy is nearly invariant (0.792 → 0.799) across a 19× write range (0.24 → 4.59M P/E), suggesting that most plasticity is redundant wear. Table 2 isolates the controller: in this ample-endurance regime the break-even gate and wear-leveling are inactive, consistent with the Remark.

Table 1: Six of seven baselines vs. SC-LACL on three drift streams (mean, 5 seeds); PRICE, the seventh, ties SC-LACL exactly here. Accuracy is per dataset; Dev (devices over the horizon), Writes (lifetime P/E, M) and tCLDP (lower is better) are for ELEC2.

	Accuracy					
Method	ELEC2	Cover	SEA	Dev	Writes	tCLDP
Static	0.66	0.75	0.82	**1**	**0.0**	1.53
Naive CL	0.80	0.86	0.86	2	17.1	2.50
Periodic	0.73	0.78	0.84	**1**	**0.8**	1.37
EWC	0.79	0.80	0.86	2	17.1	2.53
Exp. Replay	0.79	0.85	0.86	2	17.1	2.54
LwF	**0.81**	0.86	**0.87**	2	17.1	2.48
SC-LACL	0.80	**0.87**	0.87	**1**	**0.8**	**1.26**

Table 2: Ablations on ELEC2 (mean, 5 seeds); on/off columns compare the named mechanism enabled vs. disabled.

Ablation	Acc	Writes (M)	tCLDP
sparsity k=0.05	0.792	0.24	1.265
sparsity k=0.18	0.795	0.84	1.260
sparsity k=1.0 (dense)	0.799	4.59	1.254
wear-level. on/off	0.795/0.796	0.84/0.83	1.260/1.258
break-even on/off	0.795/0.795	0.84/0.84	1.260/1.260

Decomposing the gain. In this ample-endurance regime, static sparsity alone recovers much of the write reduction (k=0.05 vs. dense: 0.007 accuracy at 19× fewer writes; of ELEC2's 20× reduction, top-k gives 5.5× and drift-scaled epochs 3.7×, 17.1 → 4.59M); the controller's distinct contribution is *binding* once endurance is severely scarce.

PRICE: a static shadow-price baseline. We test this directly with **PRICE**, a new baseline identical to SC-LACL's top-k wear-leveled update except that the adaptive queues Q_E, Q_Z in the gate (Prop. 3) are replaced by constants — calibrated once from SC-LACL's own long-run average on ELEC2, then frozen — in the spirit of treating flash endurance as a fixed-price depreciating asset [9]. Because SC-LACL's wear queue stays empty in that regime, the calibrated wear price is zero, so PRICE never penalizes wear. In Table 1's regime and under moderately tight endurance (E_{cyc}=2.5×10^3), PRICE is numerically identical to SC-LACL on all three streams: the frozen price never needs to move, so adaptivity adds nothing there. The two diverge only far outside PRICE's calibration point: under severe scarcity (E_{cyc}=200, V=2 for both, ELEC2, 5 seeds), PRICE keeps higher accuracy (0.795 vs. 0.756) but needs 4.6× more devices (23 vs. 5) and a 4.4× worse tCLDP (28.97 vs. 6.62), because without a wear price it never throttles writes. Adaptivity's value is real, but conditional: it is invisible until deployment drifts far enough from the calibration regime.

Regime analysis. On ELEC2 (5 seeds), under *tight endurance* (E_{cyc}=2.5×10^3), all SC-LACL variants need two devices (the densely updated output-bias cells wear out first, while Q_Z tracks mean wear) but use 5.5× fewer writes than dense updating; under *scarce harvest*

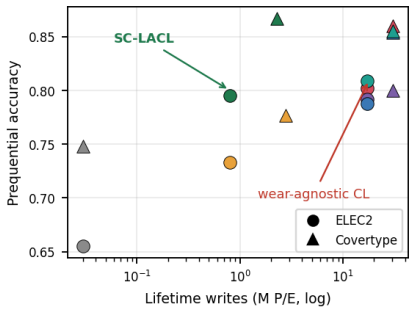


Figure 3: Accuracy vs. lifetime writes (log axis): SC-LACL reaches near-top accuracy at 13–30× fewer writes than wear-agnostic CL. Static (0 writes) is shown at the axis floor.

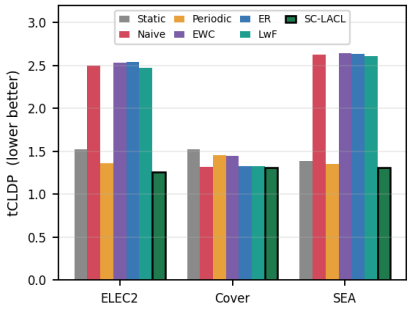


Figure 4: SC-LACL attains the best tCLDP on all three streams.

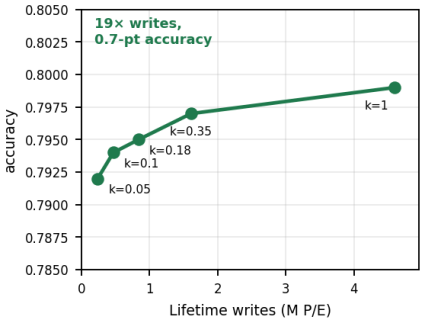


Figure 5: Accuracy is nearly flat across a 19× write range, suggesting most plasticity is redundant wear.

(mean daily harvest 0.45 vs. 0.95 nominal), low write volume gives 12.6× fewer brownout days than naive CL (126 → 10.0) and higher serviceable accuracy (0.787 vs. 0.695).

7 Research Agenda

A roadmap toward community practice: (near-term) benchmarks should log endurance traces, per-cell P/E counters, energy traces, and replacement-carbon factors alongside accuracy, calibrated by accelerated wear tests; (mid-term) lifetime-aware NAS should score candidates by accuracy, latency, write footprint, survival probability, and tCLDP, rejecting architectures that merely shift cost into NVM wear; (long-term) MLPerf Tiny [2] and related suites could add tCLDP leaderboards and uncertainty-aware embodied-carbon models [11, 38], making lifetime and carbon accounting first-class review artifacts.

8 Conclusion

Sustainable edge AI cannot be achieved by optimizing operational energy alone; embodied carbon dominates, is amortized over lifetime, and unmanaged on-device learning spends that lifetime [30]. The central shift is to model learning algorithms as physical agents whose updates consume endurance, embodied carbon, and replacement opportunities. By making lifetime endogenous, SC-LACL turns the stability–plasticity–endurance trilemma into a tractable control problem. The evidence is simulation-based, so hardware lifetime validation remains next; the broader agenda is an edge-AI design discipline that optimizes accuracy, energy, endurance, and embodied carbon together.

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